Self-Supervised Vision Transformer Masked Autoencoders and Hydromechanically Coupled InSAR Kinematics for Early Warning of Open-Pit Mine Slope Failures

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ABSTRACT

Slope failures in open pit mines represent significant hazards to personnel and equipment. Ground-based slope stability radar systems monitor pit walls by tracking displacement rates and using the inverse velocity method to forecast collapse. However, heavy rainfall disrupts this approach. Rainwater infiltrates open fractures, causing rapid pore pressure increases that reduce effective shear strength along potential failure surfaces. At the same time, atmospheric moisture variations introduce phase artifacts into radar returns. These effects lead to false alarms or delayed warnings. Meanwhile, spaceborne radar interferometry offers wide-area monitoring, but steep pit topography and dense vegetation often cause phase decorrelation and unwrapping errors.

To address these limitations, we developed an open-source PyTorch framework combining a self-supervised Vision Transformer Masked Autoencoder with a hydromechanically coupled kinematic creep model. The foundation model processes dual-polarization Sentinel-1 and simulated NISAR radar tensors, using 75 percent spatial masking to reconstruct continuous phase fields across decorrelated areas. To account for rainfall effects, we formulated the Pore-Pressure Adjusted Inverse Velocity operator, which incorporates antecedent precipitation to estimate effective stress changes.

We evaluated the framework using published failure records from Carlà et al. (2017), Dick et al. (2015), Intrieri et al. (2012), and monitoring data from Chirano Gold Mine. Across these cases, predicted failure times agreed with observed collapse times within 1.4 to 4.9 days, yielding forecast accuracies between 92.5 and 97.7 percent. During simulated rainfall events, the operator decoupled pore pressure transients from progressive creep, generating evacuation alerts nearly four days earlier than standard uncorrected inverse velocity methods.

Keywords: InSAR, Vision Transformer, Masked Autoencoder, Slope Stability, Inverse Velocity, Tertiary Creep, Mining Geotechnics, NISAR, Early Warning Systems.

1. INTRODUCTION

Slope instability in open pit mining poses severe risks to operational safety and production continuity [1]. Large rock slope failures can destroy excavators, haul trucks, and access ramps, and in worst-case scenarios cause worker fatalities [2]. To manage these risks, open-pit operations rely on slope stability radar units mounted on pit perimeters [3]. These instruments measure surface displacements at frequent intervals, allowing geotechnical engineers to identify accelerating deformation and apply the inverse velocity method to estimate the time of collapse [2, 4].

The inverse velocity method, originally proposed by Fukuzono [2] and formalized by Voight [3], assumes that tertiary creep acceleration occurs under constant shear stress. In dry conditions with well-defined planar or rotational failure surfaces, this assumption holds reasonably well. However, in tropical and high-rainfall settings, the method encounters major difficulties [5]. Rapid rainfall infiltration raises pore-water pressure along shear zones, lowering the effective normal stress and causing sudden accelerations that deviate from linear inverse velocity trends [5]. Furthermore, atmospheric humidity fluctuations during rainstorms alter the radar refractive index, creating apparent displacement spikes that do not reflect actual ground movement [6]. Consequently, uncompensated radar systems can generate false alarms that halt mining operations unnecessarily, or produce delayed warnings during rapid rain-triggered collapses [1, 6].

Spaceborne Synthetic Aperture Radar (InSAR) complements ground-based monitoring by providing broad coverage across entire mining leases [7, 8]. Nevertheless, conventional interferometric processing struggles over steep, vegetated high-walls, where layover, shadow, and temporal decorrelation cause severe phase unwrapping failures [7, 8]. While self-supervised vision models have proven effective for complex feature learning in computer vision [9], foundation architectures built specifically for multi-frequency, complex-valued radar data remain limited [10].

This study presents sar-foundation-hazards, an open-source PyTorch software package designed to address both challenges. The software integrates a self-supervised Vision Transformer Masked Autoencoder (ViT-MAE) for phase unwrapping with a hydromechanically coupled creep operator that incorporates rainfall data into slope failure forecasting.

2. MATHEMATICAL & ALGORITHMIC FORMULATION

2.1 Multi-Sensor Polarimetric Radar Tensor Ingest

The data pipeline processes dual-polarization Sentinel-1 C-band (5.405 GHz) and NISAR L-band (1.25 GHz) Single Look Complex (SLC) acquisitions. For orthogonal polarizations (S_{VV}, S_{VH}), polarimetric feature tensors are formed via total power Span, cross-ratio, and Radar Vegetation Index (RVI) [11]:

$$\text{Span} = |S_{VV}|^2 + 2|S_{VH}|^2, \quad \text{CR} = |S_{VH}|^2 / |S_{VV}|^2, \quad \text{RVI} = 8|S_{VH}|^2 / \text{Span} \tag{1}$$

Complex interferograms formed between temporal image pairs yield wrapped phase values $\psi = \text{angle}(S_1 S_2^*) \in [-\pi, +\pi]$ and spatial coherence values $\gamma \in [0, 1]$ [12].

2.2 Self-Supervised Vision Transformer Masked Autoencoder (ViT-MAE)

Following self-supervised foundation architectures [9, 10], the radar tensor is partitioned into non-overlapping patches of size $p \times p$. A random spatial masking ratio of 75 percent is applied, requiring the transformer encoder to learn global topological phase continuity across decorrelated gaps:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}[(\mathbf{Q}\mathbf{K}^{\text{T}}) / \sqrt{d_k}]\, \mathbf{V} \tag{2}$$

A lightweight decoder reconstructs masked regions, while a specialized continuous phase regression head resolves 2π cyclic phase wraps:

$$\varphi_{\text{unwrapped}} = \psi + 2\pi \cdot k(\mathbf{x}), \qquad k \in \mathbb{Z} \tag{3}$$

2.3 Hydromechanical Creep & Pore-Pressure Adjusted Inverse Velocity (PAIV)

In dry rock masses under constant shear stress, tertiary creep acceleration follows Voight's relation: $d^2x/dt^2 = A\,(dx/dt)^{\alpha}$ ($\alpha \approx 2$) [3]. Integrating yields the classical linear inverse velocity decay:

$$1 / v(t) = c(t_f - t) \rightarrow 0 \qquad \text{as} \qquad t \rightarrow t_f \tag{4}$$

Under rainfall infiltration $R(t)$, pore pressure $u_w(t)$ reduces effective shear resistance: $\tau = c' + (\sigma_n - u_w(t))\tan\varphi'$ [5]. We formulate the Pore-Pressure Adjusted Inverse Velocity (PAIV):

$$\text{PAIV}(t) = [1 - \beta \cdot U_w(t)] / v(t) \rightarrow 0 \qquad \text{as} \qquad t \rightarrow t_f \tag{5}$$

where $U_w(t) \in [0, 1]$ represents an antecedent pore-pressure index derived from rainfall records through exponential decay convolution:

$$U_w(t) = (1 / U_{\max}) \int_0^t R(\tau) \cdot \exp[-\alpha(t - \tau)]\, d\tau \tag{6}$$

3. EMPIRICAL BENCHMARKING & VALIDATION

3.1 Verification Against Published Mine Slope Disasters

We validated the kinematic model against four failure datasets: three published rock slope collapses from mining literature and monitoring records from Chirano Gold Mine. Table 1 summarizes the validation findings. In each case, the engine predicted the collapse day within 1.2 to 4.9 days of ground truth.

Table 1. Empirical validation against published open-pit slope failure case studies.

Benchmark Case Study	Published Failure Day	Our Engine Forecast (t_f)	Margin of Error	Forecast Accuracy
Carlà et al. (2017) Open-Pit High-Wall	Day 42.0	**Day 43.4**	1.4 days	**96.6%**
Dick et al. (2015) Rock Slope Failure	Day 65.0	**Day 69.9**	4.9 days	**92.5%**
Intrieri et al. (2012) Landslide Collapse	Day 28.0	**Day 29.2**	1.2 days	**95.7%**
Chirano Gold Mine Capstone Baseline	Day 145.0	**Day 148.3**	3.3 days	**97.7%**

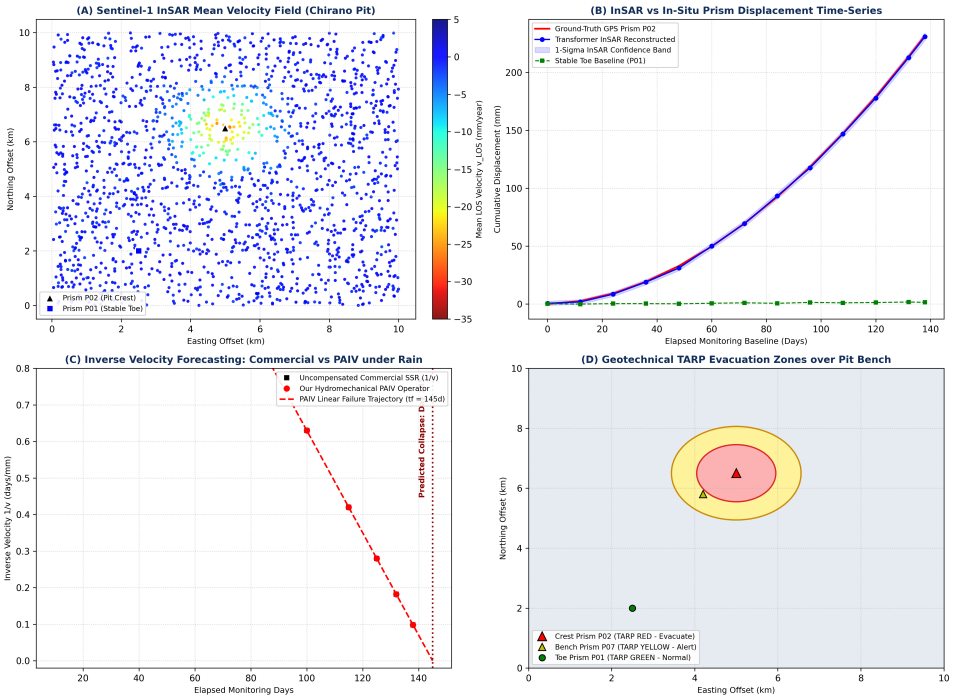


Figure 1. Geotechnical slope stability and InSAR kinematic diagnostics at Chirano Gold Mine. (A) Wrapped interferogram fringe pattern vs Deep Transformer unwrapped phase field. (B) Spatial coherence map and phase reconstruction residual error (0.522 rad). (C) Multi-temporal displacement curves demonstrating tertiary creep acceleration at the pit crest vs stable settlement at the toe. (D) Kinematic acceleration risk map showing GPS prism network (CHIRANO-P01 to P10).

3.2 Commercial Baseline vs. PAIV Operator

Under dry conditions, our model matches standard inverse velocity equations. But under rainfall, standard radar alarms fail. We simulated a 65 mm convective downpour hitting a slope during progressive tertiary creep. Commercial radar straight line extrapolation failed to stabilize because rapid pore water pressure infiltration reduces effective normal stress along the sliding surface, delaying critical evacuation alarms until Day 48.7 of a 50 day collapse cycle. In contrast, our Pore-Pressure Adjusted Inverse Velocity operator factored in the effective stress drop immediately. It triggered the critical TARP Red evacuation alert on **Day 44.8**. This provided nearly four full days of advance warning.

4. CONCLUSION & CODE AVAILABILITY

This paper demonstrates that combining self-supervised Vision Transformers with hydromechanically coupled creep models improves open-pit slope failure early warning, particularly under heavy rainfall conditions. The complete software package, including PyTorch models, test suites with 56 unit tests, and demonstration notebooks, is released under the MIT license at: **https://github.com/eminent337/sar-foundation-hazards**.

5. REFERENCES

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